

STA302 — Methods of Data Analysis

Course Notes

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Week 1: Course Overview and Simple Linear Regression

Learning Objectives

Basic Distinguish population vs. sample quantities, random variables vs. fixed values §; calculate LS coefficient estimates §2; interpret estimated coefficients in context §1.

Adv. Explain the LS process to a broad audience; derive LS estimates for various model specifications; apply SLR in R (`lm()`).

Regression Basics

Definition - Regression

$Y \approx f(X_1, \dots, X_p)$, $p \geq 1$ predictors. Y = response; X_i = predictors/covariates/features.

- **Simple** ($p = 1$) vs **Multiple** ($p > 1$).
- **Parametric**: f has a fixed parametric form, e.g. $\beta_0 + \beta_1 X$, $\beta_0 + \beta_1 X + \beta_2 X^2$. Else **non-parametric** (kNN, trees, neural nets).

The SLR Model

Definition - SLR Model

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad i = 1, \dots, n, \quad E[\varepsilon_i] = 0 \quad \forall i$$

- β_0, β_1 : fixed, unknown parameters — *same* for every i .
- ε_i ($= e_i = \epsilon_i$): random, zero-mean — *differs* by i .

Was: $y_i \stackrel{iid}{\sim} N(\mu, \sigma^2)$. Now: $y_i | x_i = \beta_0 + \beta_1 x_i + \varepsilon_i$.

Functional ($y_i = \beta_0 + \beta_1 x_i$, no error) vs **statistical** relationship ($y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$); $\varepsilon_i = y_i - (\beta_0 + \beta_1 x_i)$ = residual.

“**Linear**” = linear in β_0, β_1 , not in X .

- OK (still linear): $\beta_0 + \beta_1 x_i^2$, $\beta_0 + \beta_1 \sin x_i$, $\beta_0 + \beta_1 \mathbb{I}(\cdot)$ — free to transform X .
- NOT linear: $\log(\beta_0 + \beta_1 x_i + \varepsilon_i)$, $\beta_0 e^{\beta_1 x_i} + \varepsilon_i$ — *don't transform Y* .

Always condition on X : $E[Y | X] = \beta_0 + \beta_1 X$ (abbreviated $E[Y]$, $\text{Var}[Y]$). *$E[Y]$ alone is meaningless — need e.g. $E[Y | X = 5]$.*

Notation: sample (x_i, y_i) = realizations of (X, Y) ; population $Y = \beta_0 + \beta_1 X + \varepsilon$.

Fitted Model & Predictions

Definition - Fitted SLR

$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X$, where $\hat{\beta}_0 \rightarrow \beta_0$, $\hat{\beta}_1 \rightarrow \beta_1$, $\hat{Y} \rightarrow E[Y | X]$.

Example 1 Credit Score										
Income (X , \$100k)	1.3	1.1	0.8	1.2	1.4	0.9	0.9	1.4	1.2	1.0
Credit Score (Y)	756	728	635	599	760	722	743	726	694	726

`lm(creditScore ~ income)`: $\hat{\beta}_0 = 632.38$, $\hat{\beta}_1 = 68.32$

$$\hat{Y} = 632.38 + 68.32X$$

Interpretation:

- $\hat{\beta}_0$ = avg. Y when $X = 0$ *avg. credit score, \$0 income*
- $\hat{\beta}_1$ = avg. change in Y per unit increase in X *per \$100k income*
- Prediction: $\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X$, e.g. $X = 1.05 \Rightarrow \hat{Y} = 704.1$

Validity of predictions — two independent checks:

- Statistical:** is X inside the training range? (else **extrapolation** — training range was $X \in [0.8, 1.4]$)
- Subject-matter:** do $X, \hat{Y}$ make sense? (credit score $\notin [300, 900]$, or negative income $\Rightarrow$ invalid)

Least Squares Estimation

$$RSS = \sum_{i=1}^n \hat{\varepsilon}_i^2 = \sum_{i=1}^n (y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_i))^2 \quad \text{— minimize over } \hat{\beta}_0, \hat{\beta}_1$$

Why square: no +/− cancellation, differentiable, penalizes outliers more; geometrically $\hat{Y}$ = orthogonal projection of Y onto model space.

Identities:

$$\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) = \sum_{i=1}^n x_i y_i - n\bar{x}\bar{y}, \quad \sum_{i=1}^n (x_i - \bar{x})^2 = \sum_{i=1}^n x_i^2 - n\bar{x}^2$$

Proposition 2 LS Estimators.

$$\hat{\beta}_1 = \frac{SP_{xy}}{SS_{xx}} = \frac{\sum x_i y_i - n\bar{x}\bar{y}}{\sum x_i^2 - n\bar{x}^2} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} = \frac{\text{Cov}(X, Y)}{\text{Var}(X)}, \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

SS_{xx} = “sum of squares for x ”; SP_{xy} = “sum of products between x, y ”. No relationship $\Rightarrow \hat{\beta}_1 = 0$, $\hat{\beta}_0 = \bar{y}$.

Derivation. $L(\beta_0, \beta_1) = \sum (y_i - \beta_0 - \beta_1 x_i)^2$.

$$\frac{\partial L}{\partial \beta_0} = 0 \Rightarrow \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

$$\frac{\partial L}{\partial \beta_1} = 0 \Rightarrow \sum x_i y_i = \hat{\beta}_0 \sum x_i + \hat{\beta}_1 \sum x_i^2$$

Sub. $\hat{\beta}_0$: $\sum x_i y_i - n\bar{x}\bar{y} = \hat{\beta}_1(\sum x_i^2 - n\bar{x}^2) \Rightarrow \hat{\beta}_1$ as above. □

Formulas only valid for 1 untransformed predictor.

Example 3 Deriving for a Different Specification: No Intercept

$y_i = \beta_1 x_i + \varepsilon_i \Rightarrow L(\beta_1) = \sum (y_i - \beta_1 x_i)^2$. $\frac{\partial L}{\partial \beta_1} = -2 \sum x_i (y_i - \beta_1 x_i) = 0 \Rightarrow \hat{\beta}_1 = \frac{\sum x_i y_i}{\sum x_i^2}$. *Same method every time: write L , differentiate, set = 0, solve.*

Proposition 4 Properties, if β_0 in model.

$$\sum_{i=1}^n \hat{\varepsilon}_i = 0, \quad \sum_{i=1}^n x_i \hat{\varepsilon}_i = 0$$

2nd property: X and residuals uncorrelated (regardless of β_0). Good fit = no pattern in residuals vs. X .

Proof. Sub $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$ into both (and $\hat{\beta}_1 = SP_{xy}/SS_{xx}$ into the 2nd). □

Week 2: Multiple Linear Regression (MLR) Models

Learning Objectives

Basic Write an MLR model in matrix form §; compute LS estimates via $(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$ §8; interpret coefficients, indicators, and interactions §.

Adv. Derive $\hat{\beta}$ via matrix calculus; encode categorical predictors with dummy variables; recognize interaction effects from non-parallel slopes.

Matrices (Review)

Definition - Matrix

A rectangular array of elements in n rows and p columns (*dimension/size* $n \times p$): n = number of rows (1 per *observation*), p = number of columns (1 per *explanatory variable*). *This is exactly tidy data format: 1 row per observation, 1 column per variable.*

$$\mathbf{A} = (a_{ij})_{i=1,\dots,n; j=1,\dots,p} = \begin{pmatrix} a_{11} & \cdots & a_{1p} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{np} \end{pmatrix}$$

Square: $n = p$. **Column vector:** $p = 1$. **Row vector:** $n = 1$.

Definition - Transpose, Symmetric

If $\mathbf{A}$ is $n \times p$, its **transpose** $\mathbf{A}'$ (or $\mathbf{A}^T$) is $p \times n$ (rows $\leftrightarrow$ columns). $\mathbf{A}$ is **symmetric** if $\mathbf{A} = \mathbf{A}'$ (only possible if square).

Example 5 Design Matrix and $\mathbf{X}'\mathbf{X}$

$$\mathbf{Y} = \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} \text{ vector of responses, } \quad \mathbf{X} = \begin{pmatrix} 1 & x_1 \\ 1 & x_2 \\ 1 & x_3 \end{pmatrix} \text{ always a column of 1's}$$

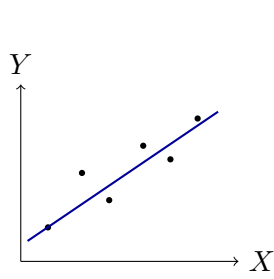
$$\mathbf{X}'\mathbf{X} = \begin{pmatrix} 1 & 1 & 1 \\ x_1 & x_2 & x_3 \end{pmatrix} \begin{pmatrix} 1 & x_1 \\ 1 & x_2 \\ 1 & x_3 \end{pmatrix} = \begin{pmatrix} 3 & \sum x_i \\ \sum x_i & \sum x_i^2 \end{pmatrix}$$

$\mathbf{X}'\mathbf{X}$ is always square and symmetric.

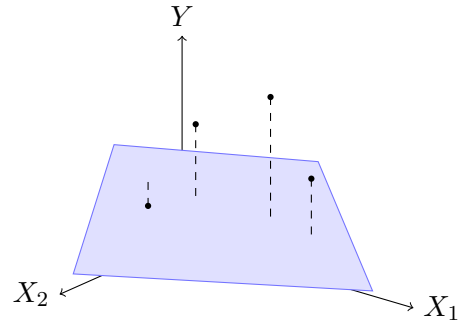
The MLR Model

Definition - Simple vs. Multiple Regression

$p = 1$: **Simple** Regression. $p > 1$: **Multiple** Regression, $f : \mathbb{R}^p \rightarrow \mathbb{R}$ — the more general form (SLR is MLR with $p = 1$).



1 predictor: line of best fit



2 predictors: plane in $\mathbb{R}^3$

$p > 2$ predictors: hyperplane in $\mathbb{R}^{p+1}$ (can't visualize directly)

Definition - MLR Model

For individual $i = 1, \dots, n$:

$$Y_i = \beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip} + \epsilon_i = \mathbf{x}'_i \boldsymbol{\beta} + \epsilon_i, \quad \epsilon_i \stackrel{iid}{\sim} N(0, \sigma^2)$$

where $\mathbf{x}'_i = (1, x_{i1}, \dots, x_{ip})$ is treated as fixed/known. Stacking over i : $\boxed{\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}}$,

$$\mathbf{X} = \begin{pmatrix} \mathbf{x}'_1 \\ \vdots \\ \mathbf{x}'_n \end{pmatrix} = \begin{pmatrix} 1 & x_{11} & \cdots & x_{1p} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & \cdots & x_{np} \end{pmatrix}, \quad \boldsymbol{\beta} = \begin{pmatrix} \beta_0 \\ \vdots \\ \beta_p \end{pmatrix}, \quad \boldsymbol{\epsilon} = \begin{pmatrix} \epsilon_1 \\ \vdots \\ \epsilon_n \end{pmatrix} \sim N(\mathbf{0}_n, \sigma^2 \mathbf{I}_n)$$

$\sigma^2 \mathbf{I}_n$: multivariate normal, diagonal covariance $\text{diag}(\sigma^2, \dots, \sigma^2)$ — same σ^2 , i.i.d. as in SLR.
 $E[Y | \mathbf{X} = \mathbf{x}] = \beta_0 + \beta_1 x_1 + \dots + \beta_p x_p$, $\text{Var}[Y | \mathbf{X} = \mathbf{x}] = \sigma^2$ (again abbreviated $E[Y]$, $\text{Var}[Y]$, always conditioning on $\mathbf{X}$).

$\mathbf{x}$ is $(p+1)$ -dimensional: 1 intercept + p predictors. (Some sources instead say p entries total, $p-1$ predictors — check convention.)

Remark (Design Matrix Conventions). • First column is **always** 1's (intercept).

- Must be numeric — categorical predictors need **indicator (dummy) variables**, a.k.a. *one-hot encoding*.
- Each observation must fall in exactly 1 category.

Example 6 Encoding a Categorical Predictor

2 levels (e.g. transmission, automatic/manual): 1 indicator suffices, $X = \mathbb{I}(\text{manual})$.

c levels (e.g. colour Blue/Red/Yellow): pick 1 **baseline** (in R: first level alphanumerically), need $c-1$ indicators. $X_2 = \mathbb{I}(\text{Red})$, $X_3 = \mathbb{I}(\text{Yellow})$; both 0 $\Rightarrow$ Blue.

Example 7 Dimension of $\mathbf{X}$

7 numeric predictors, $n = 45$: $\mathbf{X}$ is 45×8 (1 intercept + 7 predictors).

1 continuous predictor + 1 categorical predictor with 10 levels, $n = 100$: $\mathbf{X}$ is 100×11 (1 intercept + 1 continuous + 9 dummy columns). *Want p small relative to n — helps ensure $\mathbf{X}'\mathbf{X}$ is invertible.*

Discrete predictor taking values 1–100: continuous or categorical? Ask: can the levels be ordered? Is the difference between adjacent levels always the same (e.g. $2 - 1 \stackrel{?}{=} 100 - 99$)? If yes to both, treat as continuous.

Estimation via Least Squares

Definition - Residual Sum of Squares

$$RSS(\hat{\beta}) = \sum_{i=1}^n \hat{\epsilon}_i^2 = \sum_{i=1}^n (Y_i - \mathbf{x}_i' \hat{\beta})^2 = (\mathbf{Y} - \mathbf{X} \hat{\beta})' (\mathbf{Y} - \mathbf{X} \hat{\beta}) = (\mathbf{Y} - \hat{\mathbf{Y}})' (\mathbf{Y} - \hat{\mathbf{Y}})$$

Same LS procedure as SLR: (1) set up the estimating equation, (2) differentiate w.r.t. β and set to $\mathbf{0}$, (3) solve for $\hat{\beta}$, (4) unique minimizer iff $\mathbf{X}'\mathbf{X}$ is invertible (*full rank*) — see *multicollinearity*.

Matrix calculus (Renchner & Schaalje §2.14): for constants $\mathbf{a}$, $\partial(\mathbf{a}'\mathbf{x})/\partial\mathbf{x} = \mathbf{a}$; for symmetric $\mathbf{A}$, $\partial(\mathbf{x}'\mathbf{A}\mathbf{x})/\partial\mathbf{x} = 2\mathbf{A}\mathbf{x}$; and $(\mathbf{A}\mathbf{B})' = \mathbf{B}'\mathbf{A}'$.

Proposition 8 LS Estimator.

Assuming $(\mathbf{X}'\mathbf{X})^{-1}$ exists,

$$\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}'\mathbf{Y}$$

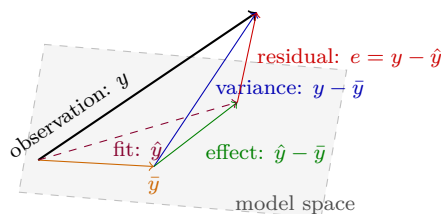
In practice, component matrices are easy to build from sums; only the inverse needs software.

$$\mathbf{X}'\mathbf{X} = \begin{pmatrix} n & \sum x_{i1} & \cdots & \sum x_{ip} \\ \sum x_{i1} & \sum x_{i1}^2 & \cdots & \sum x_{i1}x_{ip} \\ \vdots & \vdots & \ddots & \vdots \\ \sum x_{ip} & \sum x_{i1}x_{ip} & \cdots & \sum x_{ip}^2 \end{pmatrix}, \quad \mathbf{X}'\mathbf{Y} = \begin{pmatrix} \sum y_i \\ \sum x_{i1}y_i \\ \vdots \\ \sum x_{ip}y_i \end{pmatrix}$$

Definition - Hat Matrix

$\hat{\mathbf{Y}} = \mathbf{X}\hat{\beta} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y} = \mathbf{H}\mathbf{Y}$, where $\mathbf{H} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$ is the **hat/projection/influence** matrix: *idempotent* ($\mathbf{H}^2 = \mathbf{H}$), and projects $\mathbf{Y}$ onto $\hat{\mathbf{Y}}$ (*model space*).

- Diagonal entries $h_{ii} = \mathbf{x}_i'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_i$ are **leverages**: high leverage $\Rightarrow$ *outlier* in X , potential high influence on $\hat{\beta}$ if deleted.
- Residuals: $\hat{\epsilon} = \mathbf{Y} - \hat{\mathbf{Y}} = (\mathbf{I}_n - \mathbf{H})\mathbf{Y} = \mathbf{M}\mathbf{Y}$, where $\mathbf{M} = \mathbf{I}_n - \mathbf{H}$ is the **residual maker** ($\hat{\epsilon} = \mathbf{M}\mathbf{Y}$) or **annihilator** matrix ($\mathbf{M}\mathbf{X} = \mathbf{0}$).



$\hat{y}$ is the orthogonal projection of y onto the model space; $e \perp$ model space.

Application Example

Example 9 mtcars Dataset

32 automobiles (1973–74 models); using `mpg` (response), `disp`, `hp`, `wt` (predictors):

$$mpg_i = \beta_0 + \beta_1 disp_i + \beta_2 hp_i + \beta_3 wt_i + \epsilon_i$$

Fitted: $\widehat{mpg} = 37 - 0.009 disp - 0.03 hp - 3.8 wt$. *disp not significant ($p = 0.93$); hp, wt significant ($p = 0.011, 0.001$); $R^2 = 0.827$.*

Example 10 Rencher & Schaalje Ex. 7.2 — Worked by Hand

12 observations of Y on X_1, X_2 give $\sum y_i = 90$, $\sum x_{i1} = 52$, $\sum x_{i2} = 102$, $\sum x_{i1}^2 = 296$, $\sum x_{i2}^2 = 1004$, $\sum x_{i1}x_{i2} = 536$, $\sum x_{i1}y_i = 482$, $\sum x_{i2}y_i = 872$, so

$$\mathbf{X}'\mathbf{X} = \begin{pmatrix} 12 & 52 & 102 \\ 52 & 296 & 536 \\ 102 & 536 & 1004 \end{pmatrix}, \quad \mathbf{X}'\mathbf{Y} = \begin{pmatrix} 90 \\ 482 \\ 872 \end{pmatrix}$$

Inverting (software) and multiplying:

$$\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y} \approx \begin{pmatrix} 5.375 \\ 3.012 \\ -1.285 \end{pmatrix} \Rightarrow \hat{y}_i = 5.375 + 3.012x_{i1} - 1.285x_{i2}$$

Example 11 Prediction

Using Ex. 10 at $X_1 = 5$, $X_2 = 9$:

$$\hat{y} = 5.375 + 3.012(5) - 1.285(9) = 8.87$$

Interpretation: the mean response when $X_1 = 5$, $X_2 = 9$ is 8.87.

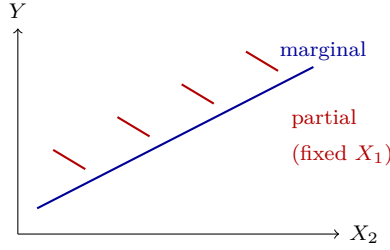
Interpreting Coefficients

Definition - Coefficient Interpretation

$$\hat{Y} = \hat{E}[Y \mid \mathbf{X} = \mathbf{x}] = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \cdots + \hat{\beta}_p x_p = \mathbf{x}'\hat{\beta}.$$

- $\hat{\beta}_0$: estimated avg. Y when *all* predictors are 0.
- $\hat{\beta}_j$ ($j \geq 1$): estimated avg. change in Y per 1-unit increase in x_j , **holding all other predictors fixed**.

Remark (Conditional Nature of MLR). Coefficients are only valid *holding other predictors fixed* — an SLR fit on one predictor alone can differ drastically (even in sign) from its MLR coefficient. In Ex. 10, SLR gives $\hat{y} = 1.86 + 1.30x_1$ and $\hat{y} = 0.86 + 0.78x_2$ (both *positive* slopes), but jointly $\hat{\beta}_2 = -1.285$ (*negative*) — because MLR's $\hat{\beta}_2$ is the effect of X_2 *within* a fixed value of X_1 (a *partial regression*), not *marginally*. *Ignoring a correlated predictor can flip the apparent sign of an effect.*



Marginal slope (blue, ignoring X_1) vs. partial slopes (red, at fixed X_1) — opposite sign here.

Example 12 Interpreting an Indicator Variable

Add `am` ($\mathbb{I}(am = 1) = \text{manual}$) to the `mtcars` model:

$$\widehat{mpg} = \hat{\beta}_0 + \hat{\beta}_1 disp + \hat{\beta}_2 hp + \hat{\beta}_3 wt + \hat{\beta}_4 \mathbb{I}(am = 1), \quad \hat{\beta}_4 = 2.16$$

$\hat{\beta}_4$ = estimated avg. difference in *mpg* between manual and automatic transmission, holding *disp*, *hp*, *wt* fixed: the intercept is $\hat{\beta}_0$ when $am = 0$, and $\hat{\beta}_0 + \hat{\beta}_4$ when $am = 1$ — a parallel vertical shift, same slopes.

Definition - Interaction

An **interaction effect** exists when the joint effect of two (or more) predictors differs from their separate effects; include it as a product term $x_j x_k$ (works for any combination of numeric/categorical predictors). *E.g. is the smoking–lung-cancer association the same in rural vs. urban areas? Does a renovation raise price in the suburbs but not downtown? Does drug dosage increase response in females but decrease it in males?*

“Linear” still just means linear in β : e.g. $Y = \beta_0 + \beta_1 x_1 x_2 + \beta_2 \sin(x_2) + \beta_3 x_3^2$ is a linear model.

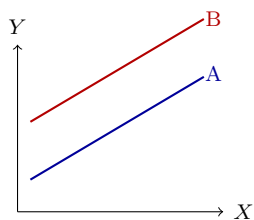
Example 13 Interaction — `mtcars`

$$\widehat{mpg} = \hat{\beta}_0 + \hat{\beta}_1 wt + \hat{\beta}_2 \mathbb{I}(am = 1) + \hat{\beta}_3 (wt \times \mathbb{I}(am = 1))$$

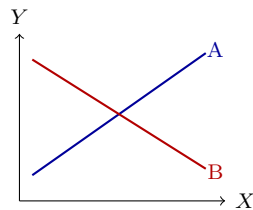
Fitted: $31.42 - 3.79wt + 14.88 \mathbb{I}(am = 1) - 5.30(wt \times \mathbb{I}(am = 1))$. Slope of *wt* is -3.79 for automatic ($am = 0$); for manual ($am = 1$) it is 5.30 more negative, i.e. -9.09 — different intercept and slope per group.

Remark (Detecting Interaction Visually). No interaction $\Rightarrow$ fitted lines/planes for each category are **parallel**. **Non-parallel** slopes (e.g. *mpg* vs. *wt* across $cyl \in \{4, 6, 8\}$) suggest an interaction (formal test later). With > 2 predictors, “lines” become planes/hyperplanes.

Plotting caution: don’t lead the viewer by extrapolating fitted lines beyond each group’s observed range — and don’t recode a numeric predictor (e.g. `cyl`) as categorical for a plot like this unless there’s a substantive reason to.



No interaction: parallel slopes



Interaction: non-parallel (crossing) slopes

A categorical predictor with $c > 2$ levels still only needs $c - 1$ dummy columns, generated automatically by software (e.g. `factor(cyl)` for `cyl` ∈ {4, 6, 8} creates 2 columns, baseline = smallest level, 4).

Example 14 Try at Home

Given the data below:

Y	X_1	X_2
-0.2	0.5	0.5
-1.1	0.8	0.7
-0.8	0.6	-0.5
-0.7	-0.2	-0.5

Write out $\mathbf{X}$ (with intercept column) and compute $\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$ by hand; check against `lm()`.